

A fitness-independent family of simultaneous amplifiers below fitness three halves

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Abstract

We study fixation from a uniformly random single mutant on finite connected loopless undirected weighted graphs under both Birth–death (Bd) and death–Birth (dB) Moran updating. We construct one explicit family, with positive rational weights independent of mutant fitness, whose fixation probability converges to $1/3$ under both rules for every fixed fitness $r > 1$. Since the corresponding complete-graph probabilities converge to $1 - 1/r$, the family strictly amplifies both rules for every fixed $1 < r < 3/2$ once the population is sufficiently large. Thus the asymptotic simultaneous-amplification threshold satisfies $R_{\text{sim}} \geq 3/2$. The proof solves the singular three-vertex module exactly, derives a marked rare-edge trace with explicit coupling error, and controls both the initial module-to-center transition and the subsequent center sweep. At $r = 3/2$ we retain the order- N^{-2} terms and prove that this same family suppresses both rules. Hence $(1, 3/2)$ is the exact amplification interval of the constructed family. We also give an exact unweighted graph on 36 vertices which refutes two stronger candidate endpoint separators: both the normalized fixation product and the balanced normalized arithmetic mean exceed one. A growing clique–pendant family strengthens this to a nonvanishing product violation, with normalized limits $32/27$ and $8/9$. Varying its leaf proportion proves that every universal convex affine separator must give Bd coefficient at most $1/3$; this upper bound is sharp among all such separators. The sharp one-third inequality itself remains open. All counterexamples here suppress dB, so no universal upper bound, and therefore no exact value of R_{sim} , is claimed.

Scope. This paper proves the lower bound $R_{\text{sim}} \geq 3/2$ and the exact interval of one explicit family. A universal upper bound and the exact value of R_{sim} remain **open**.

Author summary

An amplifier is a population structure on which a beneficial mutation is more likely to take over than in a well-mixed population of the same size. The two standard update orders—choosing birth first or death first—often favor different structures. We give one sequence of weighted graphs that works for both update orders throughout the entire fixed-fitness interval $1 < r < 3/2$. Each graph has a small central clique and many almost isolated three-vertex modules. A mutation usually starts in a module, where it has asymptotic chance $1/3$ to occupy the module. A carefully separated set of edge scales then makes an occupied module seed the center and makes an occupied center sweep the remaining modules, both with probability tending to one. Exact formulas and explicit error estimates turn this timescale picture into a proof. The construction fails at the endpoint $r = 3/2$ by a computed order- N^{-2} deficit. Whether some different family reaches or passes that endpoint remains open. An exact 36-vertex calculation included below shows, however,

that neither the normalized product nor the balanced arithmetic mean can serve as the universal endpoint separator. A growing version explains the sharp remaining affine weights: Bd can gain asymptotically two units for each unit lost by dB, making 1 : 2 the maximal possible Bd:dB weight ratio.

1 Introduction

For a fixed population size, a graph is a simultaneous amplifier at fitness r if it increases uniformly initialized fixation probability, relative to the complete graph, under both Bd and dB updating. The large-population question has a delicate order of quantifiers: one graph family must be chosen independently of r , but the population size from which amplification holds may depend on r . A previous weighted modular construction established simultaneous amplification on the interval $(1, 1.2)$ [1]. The present construction extends the rigorous lower endpoint benchmark to $3/2$.

The mechanism is a singular three-scale hierarchy. A module is a weighted triangle with one edge of weight one and two edges of weight $\delta_N = N^{-4}$. There are N^2 modules. The center is a clique of N vertices with edge weight $z_N = N^{-3}$, and every center-module edge has the much smaller rational weight $\varepsilon_N = N^{-N^3}$. The very small final scale does not itself create the bias. Rather, it separates internal absorption from intercomponent motion sufficiently strongly that an exact marked trace approximates the original chain through all polynomially many excursions.

Two comparisons drive the trace. Starting from a mutant module, the odds of fixing a mutant in the center before erasing the module grow as a positive constant times N^2 for each rule. Once the center is mutant, the chance of a resident module reversing it is exponentially small, while all N^2 modules are converted in only polynomially many trace excursions. The isolated triangle therefore determines the limiting fixation probability, and its uniformly averaged value tends to $1/3$ under both rules.

The proof also explains why the endpoint requires a separate calculation. For $r < 3/2$, a positive constant gap separates $1/3$ from the complete-graph limit. At $r = 3/2$ that gap vanishes, so we compute the first-stage failure probability, the center-start contribution, and the finite complete-graph correction through order N^{-2} . Both comparisons are then strictly negative. This establishes the exact interval for this family only.

2 Model, baselines, and the asymptotic threshold

Let $G = (V, w)$ be a finite connected loopless undirected weighted graph, $|V| = n$, with

$$w_{uv} = w_{vu} \geq 0, \quad w_{uu} = 0, \quad d_u := \sum_v w_{uv} > 0.$$

A state is the mutant set $S \subseteq V$. Mutants have fitness $r > 1$ and residents have fitness one. Put $F(S) = r|S| + n - |S|$.

Under Bd updating, first a parent is chosen proportional to fitness and then one of its replacement targets is chosen proportional to the incident edge weight. Thus, for $v \notin S$ and $v \in S$, respectively,

$$P_{\text{Bd}}(S, S \cup \{v\}) = \frac{r}{F(S)} \sum_{u \in S} \frac{w_{uv}}{d_u}, \quad (1)$$

$$P_{\text{Bd}}(S, S \setminus \{v\}) = \frac{1}{F(S)} \sum_{u \notin S} \frac{w_{uv}}{d_u}. \quad (2)$$

Under dB updating, first a uniformly chosen vertex dies and then a neighbor is chosen proportional to fitness times edge weight. Hence

$$P_{\text{dB}}(S, S \cup \{v\}) = \frac{1}{n} \frac{r \sum_{u \in S} w_{uv}}{r \sum_{u \in S} w_{uv} + \sum_{u \notin S} w_{uv}}, \quad v \notin S, \quad (3)$$

$$P_{\text{dB}}(S, S \setminus \{v\}) = \frac{1}{n} \frac{\sum_{u \notin S} w_{uv}}{r \sum_{u \in S} w_{uv} + \sum_{u \notin S} w_{uv}}, \quad v \in S. \quad (4)$$

The remaining probability is a self-loop. With $h_U(S)$ denoting fixation probability from S , these transition probabilities determine the unique solution of

$$h_U(S) = \sum_T P_U(S, T) h_U(T), \quad h_U(\emptyset) = 0, \quad h_U(V) = 1.$$

We use uniform singleton initialization:

$$\rho_U(G, r) := \frac{1}{n} \sum_{v \in V} h_U(\{v\}), \quad U \in \{\text{Bd}, \text{dB}\}.$$

2.1 Complete-graph baselines

On the unit-weight complete graph K_n , the mutant count is a birth–death chain. For Bd the adjacent down/up ratio is $1/r$, and the standard one-dimensional harmonic recurrence gives

$$\rho_{\text{Bd}}(K_n, r) = \frac{1 - r^{-1}}{1 - r^{-n}}. \quad (5)$$

For dB, let T_i^+ and T_i^- be the probabilities that the mutant count moves from i to $i + 1$ and $i - 1$. Direct substitution in (3)–(4) gives the telescoping product

$$\prod_{j=1}^i \frac{T_j^-}{T_j^+} = \frac{n - 1 + (r - 1)i}{(n - 1)r^i}.$$

Substitution into the birth–death-chain fixation formula yields

$$\rho_{\text{dB}}(K_n, r) = \frac{n - 1}{n} \frac{1 - r^{-1}}{1 - r^{-(n-1)}}. \quad (6)$$

For each fixed $r > 1$, both expressions converge to

$$q(r) := 1 - \frac{1}{r}. \quad (7)$$

2.2 The order of quantifiers

Definition 1. Let R_{sim} be the supremum of all $R > 1$ for which there exists a single family $\{G_m\}$, with $|V(G_m)| \rightarrow \infty$ and weights independent of fitness, such that

$$\forall r \in (1, R) \exists m_0(r) \forall m \geq m_0(r) : \quad \rho_{\text{Bd}}(G_m, r) > \rho_{\text{dB}}(K_{|V(G_m)|}, r)$$

and simultaneously

$$\rho_{\text{dB}}(G_m, r) > \rho_{\text{dB}}(K_{|V(G_m)|}, r).$$

The threshold $m_0(r)$ need not be uniform as $r \downarrow 1$ or as r approaches the upper endpoint. None of the limits below interchanges those quantifiers.

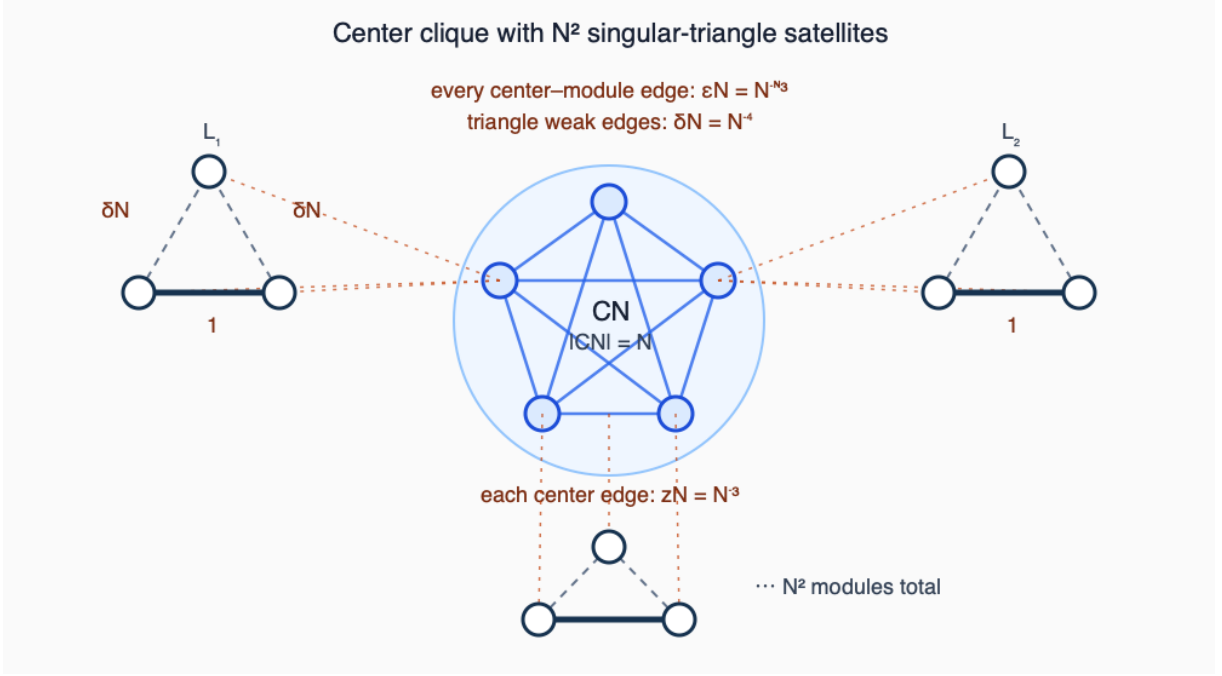


Figure 1: The center clique and three representative singular-triangle modules. Every center is joined to every module vertex by an ε_N -edge; only representative outer edges are drawn.

3 The center-triangle family and main theorem

For each integer $N \geq 3$, put

$$c_N = N, \quad M_N = N^2, \quad \delta_N = N^{-4}, \quad z_N = N^{-3}, \quad \varepsilon_N = N^{-N^3}. \quad (8)$$

The graph G_N has a center set C_N of c_N vertices and M_N disjoint three-vertex modules. In module j , with vertices A_j, B_j, D_j , set

$$w(A_j, D_j) = 1, \quad w(A_j, B_j) = w(B_j, D_j) = \delta_N. \quad (9)$$

Every pair of distinct center vertices has weight z_N . Every center vertex is joined to every module vertex by an edge of weight ε_N . There are no edges between distinct modules. Therefore

$$n_N := |V(G_N)| = N + 3N^2. \quad (10)$$

All displayed weights are positive rational numbers, G_N is connected, and the construction is independent of r .

Theorem 2 (Simultaneous amplification below three halves). *For each fixed $r > 1$,*

$$\lim_{N \rightarrow \infty} \rho_{\text{Bd}}(G_N, r) = \lim_{N \rightarrow \infty} \rho_{\text{dB}}(G_N, r) = \frac{1}{3}. \quad (11)$$

Consequently, for every fixed $1 < r < 3/2$, both strict complete-graph comparisons hold for all sufficiently large N . In particular,

$$R_{\text{sim}} \geq \frac{3}{2}.$$

For this particular family, $(1, 3/2)$ is the exact asymptotic fixed-fitness amplification interval. At the endpoint,

$$\rho_{\text{Bd}}(G_N, 3/2) - \rho_{\text{Bd}}(K_{n_N}, 3/2) = -\frac{4}{3N^2} + o(N^{-2}), \quad (12)$$

$$\rho_{\text{dB}}(G_N, 3/2) - \rho_{\text{dB}}(K_{n_N}, 3/2) = -\frac{16}{81N^2} + o(N^{-2}). \quad (13)$$

For every fixed $r > 3/2$, the family eventually suppresses both rules.

The rest of the paper proves the theorem. Equations (12)–(13) do not imply a universal obstruction at $r = 3/2$.

4 Exact calculation on one singular triangle

Write H_δ for one module with $\delta > 0$, and label the middle vertex of the two weak edges by B . Its weighted degrees are

$$d_A = d_D = 1 + \delta, \quad d_B = 2\delta. \quad (14)$$

For $U \in \{\text{Bd}, \text{dB}\}$, let $f_v^U(\delta, r)$ be fixation in H_δ from a mutant at v , and define

$$\alpha_U(\delta, r) = \frac{1}{3} \sum_{v \in \{A, B, D\}} f_v^U(\delta, r).$$

Let $\beta_v^U(\delta, r)$ be fixation of a single resident at v in an otherwise mutant module. After multiplying all fitnesses by $1/r$, this is $f_v^U(\delta, 1/r)$.

For completeness, the exact calculation uses all six nonabsorbing subsets $S \subset \{A, B, D\}$. In each such state one inserts the three edge weights in (1)–(4), removes the self-loop, and solves

$$\left(\sum_{T \neq S} P_U(S, T) \right) h_U(S) - \sum_{\emptyset \neq T \neq S, T \neq V} P_U(S, T) h_U(T) = P_U(S, V). \quad (15)$$

Reflection $A \leftrightarrow D$ reduces the six unknowns to four, although the independent verifier retains all six. Direct elimination gives the following rational expressions.

Set

$$\begin{aligned} D_B &= 9r^2 + \delta(12r^4 + 24r^3 + 15r^2 + 24r + 12) \\ &\quad + \delta^2(16r^4 + 52r^3 + 83r^2 + 52r + 16) \\ &\quad + \delta^3(4r^4 + 12r^3 + 13r^2 + 12r + 4). \end{aligned}$$

Then

$$\alpha_{\text{Bd}} = \frac{r^2\{3 + \delta(12r^2 + 12r + 5) + \delta^2(16r^2 + 36r + 21) + \delta^3(4r^2 + 8r + 3)\}}{D_B}, \quad (16)$$

and

$$\alpha_{\text{dB}} = \frac{2r\{r + 1 + \delta(3r^2 + 3r + 1) + \delta^2(5r + 1)\}}{3(r + 1)\{2r + \delta(3r^2 + r + 3) + 6\delta^2r\}}. \quad (17)$$

The reverse-direction quantities needed later are

$$X(\delta) := \sum_v \frac{1}{d_v} = \frac{2}{1 + \delta} + \frac{1}{2\delta}, \quad (18)$$

$$\begin{aligned}
Y_B(\delta, r) &:= \sum_v \beta_v^{\text{Bd}}(\delta, r) \\
&= \frac{3\{3r^2 + \delta(5r^2 + 12r + 12) + \delta^2(21r^2 + 36r + 16) + \delta^3(3r^2 + 8r + 4)\}}{D_B},
\end{aligned} \tag{19}$$

and

$$\begin{aligned}
I_D(\delta, r) &:= \sum_v \frac{\beta_v^{\text{dB}}(\delta, r)}{d_v} \\
&= \frac{2r^2 + 3r + 1 + \delta(3r^2 + 7r + 5) + \delta^2(r^2 + 8r)}{(1 + \delta)(r + 1)\{2r + \delta(3r^2 + r + 3) + 6\delta^2r\}}.
\end{aligned} \tag{20}$$

All displayed denominators have positive coefficients for $r, \delta > 0$. In particular, direct limits give

$$\begin{aligned}
\alpha_{\text{Bd}}(\delta, r) &= \frac{1}{3} + O_r(\delta), & Y_B(\delta, r) &= 1 + O_r(\delta), \\
\alpha_{\text{dB}}(\delta, r) &= \frac{1}{3} + O_r(\delta), & I_D(\delta, r) &= \frac{2r + 1}{2r} + O_r(\delta), \\
2\delta X(\delta) &= 1 + O(\delta).
\end{aligned} \tag{21}$$

For the center-to-module direction, the same solution gives

$$\sum_v f_v^{\text{Bd}} = 1 + O_r(\delta), \tag{22}$$

and exactly

$$\begin{aligned}
\sum_v \frac{f_v^{\text{dB}}}{d_v} &= \frac{r\{r^2 + 3r + 2 + \delta(5r^2 + 7r + 3) + \delta^2(8r + 1)\}}{(1 + \delta)(r + 1)\{2r + \delta(3r^2 + r + 3) + 6\delta^2r\}} \\
&= \frac{r + 2}{2} + O_r(\delta).
\end{aligned} \tag{23}$$

5 Exact center formulas

Let $Q_c^U(r)$ be fixation from one mutant in an isolated unit-weight K_c , and let $\bar{Q}_c^U(r)$ be fixation of one resident in an otherwise mutant center. The center weights are all equal, so their common scale does not affect the isolated chain. Equations (5)–(6), together with fitness inversion for the resident lineage, give

$$Q_c^{\text{Bd}}(r) = \frac{1 - r^{-1}}{1 - r^{-c}}, \quad \bar{Q}_c^{\text{Bd}}(r) = \frac{r - 1}{r^c - 1}, \tag{24}$$

$$Q_c^{\text{dB}}(r) = \frac{c - 1}{c} \frac{1 - r^{-1}}{1 - r^{-(c-1)}}, \quad \bar{Q}_c^{\text{dB}}(r) = r^{2-c} Q_c^{\text{dB}}(r). \tag{25}$$

Thus $Q_c^U(r) \rightarrow q(r)$, while $\bar{Q}_c^U(r)$ is exponentially small in c .

6 A quantitative rare-edge trace

We now justify the separation between internal absorption and outer motion. The statement is deliberately stronger than the estimate needed by the construction.

Lemma 3 (Rare-edge trace). *Fix $r > 1$. Suppose a graph on n vertices is partitioned into internally connected components of size at most s . Every component contains at least two vertices, every nonzero internal edge lies in $[a, b]$, and every outer edge has weight at most $\varepsilon \leq b$. Start with all components monomorphic except possibly one. There is a constant C_r such that, with*

$$\eta(n, s, a, b, \varepsilon) := C_r n^4 \frac{b\varepsilon}{a^2} s \left(\frac{C_r n^2 b}{a} \right)^s, \quad (26)$$

the probability, during one excursion to the next componentwise-monomorphic state, of either a second type-changing outer event before internal absorption or a discrepancy from isolated-component absorption is at most η . The first L excursions can be coupled with the marked trace chain with error at most $L\eta$.

The marked trace chooses an outer replacement at its rate in the current componentwise-monomorphic state and marks it successful with the isolated fixation probability of the introduced type in the target component.

Proof. While a component is polymorphic it contains a discordant internal edge. Under either rule, one orientation of such an edge changes type in a desired direction in one global update with probability at least

$$p = \frac{a}{C_r n^2 b}. \quad (27)$$

The probability of any type-changing outer replacement in one global update is at most $C_r n \varepsilon / a$.

From every nonabsorbing internal configuration, choose adaptively a monotone sequence of at most s replacements along a spanning tree that makes the active component monomorphic. The next s global updates realize this sequence, with harmless padding after absorption, with conditional probability at least p^s . Hence internal absorption occurs in each block of s updates with conditional probability at least p^s , and its expected duration is at most $s p^{-s}$. Multiplying by the outer-event bound controls an intervening outer replacement.

Deleting the outer edges changes a Bd neighbor-choice distribution or a dB parent-choice distribution in one update by total variation at most $C_r n \varepsilon / a$. An update-by-update coupling for the same expected duration controls the discrepancy from the isolated law. Both estimates are below the generous envelope (26) for $n \geq 2$ and $b \geq a$. A union bound gives the L -excursion statement. $\square$

For G_N , use the center as one component and each triangle as another, and take

$$n \leq 4N^2, \quad s = N, \quad a = N^{-4}, \quad b = 1, \quad \varepsilon = N^{-N^3}. \quad (28)$$

Then

$$\log \eta_N = -N^3 \log N + O_r(N \log N), \quad (29)$$

so $N^K \eta_N \rightarrow 0$ for every fixed K . The rational outer weight therefore supports all polynomial excursion bounds used below.

7 From one mutant module to a mutant center

Put

$$Z_N = (c_N - 1)z_N = (N - 1)N^{-3} \sim N^{-2}, \quad (30)$$

the internal weighted degree of a center vertex. Suppose one module is all mutant while the center and all other modules are resident. In the marked trace, the competing successful macro changes are fixation of an introduced mutant in the center and fixation of an introduced resident in the mutant module. Failed introductions return to the same macro state.

For Bd, direct summation of source-selection, target-selection, and isolated fixation factors gives the successful-rate ratio

$$A_{B,N} = \frac{(Z_N + 3M_N \varepsilon_N) r Q_{c_N}^{\text{Bd}}(r) \sum_v (d_v + c_N \varepsilon_N)^{-1}}{Y_B(\delta_N, r)} [1 + o(1)]. \quad (31)$$

Here and below v ranges over the three vertices of the active module. The $o(1)$ is covered uniformly along the coupled trace by Lemma 3. Equations (21), (24), and (30) yield

$$A_{B,N} \sim \frac{r-1}{2} \frac{Z_N}{\delta_N} \sim \frac{r-1}{2} N^2 \longrightarrow \infty. \quad (32)$$

For dB, the two successful rates themselves are

$$a_{D,N} = \frac{c_N}{n_N} \frac{3r \varepsilon_N}{Z_N + 3(M_N - 1) \varepsilon_N + 3r \varepsilon_N} Q_{c_N}^{\text{dB}}(r), \quad (33)$$

$$b_{D,N} = \frac{1}{n_N} \sum_v \frac{c_N \varepsilon_N}{r d_v + c_N \varepsilon_N} \beta_v^{\text{dB}}(\delta_N, r). \quad (34)$$

Thus

$$A_{D,N} := \frac{a_{D,N}}{b_{D,N}} \sim \frac{3r(r-1)}{Z_N I_D(\delta_N, r)} \sim \frac{6r^2(r-1)}{2r+1} N^2 \longrightarrow \infty. \quad (35)$$

For either rule, the chance that the center is seeded before the mutant module is erased is

$$\frac{A_{U,N}}{1 + A_{U,N}} + o(1), \quad (36)$$

and hence tends to one.

This stage also requires only polynomially many excursions. The exact module formulas imply that a relevant outer introduction produces a successful macro change with probability at least $c_r \delta_N = c_r N^{-4}$ for all sufficiently large N . The expected number of introductions is at most $C_r N^4$, so

$$\Pr\{\text{first stage uses more than } N^{10} \text{ introductions}\} = O_r(N^{-6}). \quad (37)$$

The two rules prefer opposite center scales. If $\delta \rightarrow 0$ and $c \rightarrow \infty$ while the total center degree Z is free, comparison of the first-stage probability with the baseline $q(r)$ gives the leading window

$$Z > \frac{6\delta}{3-2r} \quad (\text{Bd}), \quad Z < \frac{2r^2(3-2r)}{2r+1} \quad (\text{dB}). \quad (38)$$

It is nonempty precisely in the relevant direction for $1 < r < 3/2$. The choices $\delta_N \asymp N^{-4}$ and $Z_N \asymp N^{-2}$ lie strictly inside it for every fixed such r once N is large. This observation explains the constant $3/2$ for the construction, but is not a universal bound.

8 From a mutant center to global fixation

Suppose the center is mutant and R modules remain resident. At the next successful trace transition, either one resident module becomes mutant or a resident from a module fixes in the center. The factor R cancels from the bad-to-good successful-rate ratio. Equations (18), (19), and (24) give

$$B_{B,N} \leq C_r Z_N X(\delta_N) \bar{Q}_{c_N}^{\text{Bd}}(r) \leq C_r N^2 r^{-N}. \quad (39)$$

Similarly, (23), (25), and (30) give

$$B_{D,N} \leq C_r Z_N^{-1} \bar{Q}_{c_N}^{\text{dB}}(r) \leq C_r N^2 r^{2-N}. \quad (40)$$

There are $M_N = N^2$ successful module conversions, so a union bound proves that center reversal before all modules convert has probability $o(1)$.

It remains to verify that the sweep does not require too many failed introductions. Direct raw-rate comparison and the exact isolated fixation probabilities give the following lower bounds for the chance that one type-changing outer excursion produces a successful macro change:

	module-to-center stage	center-sweep stage
Bd	c_r	$c_r N^{-2}$
dB	$c_r \delta_N$	$c_r \delta_N$

The N^{-2} factor in the Bd sweep is essential: adverse module-to-center introductions occur more often than favorable center-to-module introductions, but almost always fail to fix in the mutant center. With N^2 required conversions, the expected total number of excursions is at most $C_r N^6$. Therefore

$$\Pr\{\text{sweep uses more than } N^{10} \text{ introductions}\} \leq C_r N^{-4} = o(1). \quad (41)$$

By (29), the coupling error through these excursions is $N^{10} \eta_N = o(1)$. Hence a mutant center produces global fixation with probability tending to one under both rules.

9 Uniform initialization and proof below the endpoint

The uniformly chosen initial mutant lies in the center with probability

$$h_N := \frac{N}{N + 3N^2} = \frac{1}{3N + 1} = O(N^{-1}). \quad (42)$$

Otherwise it is uniform on the three vertices of one module. Before an outer replacement occurs, the module follows its isolated chain up to $o(1)$ by Lemma 3. It fixes internally with probability

$$\alpha_U(\delta_N, r) = \frac{1}{3} + o(1).$$

Conditional on internal fixation, the preceding two sections give global fixation with probability $1 - o(1)$. Conditional on internal extinction there are no mutants. Since the center-start mass is $o(1)$, this proves (11).

For fixed $1 < r < 3/2$,

$$\frac{1}{3} - q(r) = \frac{3 - 2r}{3r} > 0. \quad (43)$$

The graph probabilities converge to $1/3$, while the exact complete-graph probabilities (5)–(6) converge to $q(r)$. Hence there exists $N_0(r)$ such that both strict finite- N comparisons hold for every $N \geq N_0(r)$. The same graph sequence works for all fixed r in the interval, so $R_{\text{sim}} \geq 3/2$.

10 Endpoint expansion and the exact interval of the family

Set $r = 3/2$ and $q = 1/3$. To identify the sign after the limiting gap closes, we first account for all approximation errors. The two Markov tails are $O(N^{-6})$ and $O(N^{-4})$, the trace coupling contributes at most $N^{10}\eta_N$, and the sweep reversal union bound is $O_r(N^4 r^{-N})$. Thus the complement of the coupled direct path has probability

$$O_r(N^{-4}) + N^{10}\eta_N + O_r(N^4 r^{-N}) = o(N^{-2}). \quad (44)$$

The same bound applies to a center start. No omitted event can therefore alter an order- N^{-2} coefficient.

Equations (16)–(21) give

$$\alpha_{\text{Bd}}(\delta_N, 3/2) = q + O(N^{-4}), \quad \alpha_{\text{dB}}(\delta_N, 3/2) = q + O(N^{-4}). \quad (45)$$

The first-stage odds sharpen to

$$A_{B,N} = \frac{1}{4}N^2(1 + o(1)), \quad A_{D,N} = \frac{27}{16}N^2(1 + o(1)). \quad (46)$$

Let $p_{U,N}$ be global fixation from a uniformly chosen singleton in one module. The marked renewal chain and (44) give

$$p_{U,N} = \alpha_U(\delta_N, 3/2) \frac{A_{U,N}}{1 + A_{U,N}} + o(N^{-2}).$$

Therefore

$$p_{B,N} = q - \frac{4}{3N^2} + o(N^{-2}), \quad p_{D,N} = q - \frac{16}{81N^2} + o(N^{-2}). \quad (47)$$

Let $q_{U,N}$ be global fixation from a uniformly chosen center singleton. The center first follows its isolated count chain, and after center fixation the sweep fails with probability $o(N^{-2})$. Consequently,

$$q_{B,N} = Q_N^{\text{Bd}}(3/2) + o(N^{-2}) = q + o(N^{-2}), \quad (48)$$

$$q_{D,N} = Q_N^{\text{dB}}(3/2) + o(N^{-2}) = q - \frac{q}{N} + o(N^{-2}). \quad (49)$$

Uniform initialization is exactly

$$\rho_U(G_N, 3/2) = (1 - h_N)p_{U,N} + h_N q_{U,N}. \quad (50)$$

Using $h_N = 1/(3N + 1)$, we obtain

$$\rho_{\text{Bd}}(G_N, 3/2) = q - \frac{4}{3N^2} + o(N^{-2}), \quad (51)$$

$$\rho_{\text{dB}}(G_N, 3/2) = q - \frac{25}{81N^2} + o(N^{-2}). \quad (52)$$

For $n_N = N + 3N^2$, the exact baselines give

$$\rho_{\text{Bd}}(K_{n_N}, 3/2) = q + o(N^{-2}), \quad \rho_{\text{dB}}(K_{n_N}, 3/2) = q - \frac{1}{9N^2} + o(N^{-2}). \quad (53)$$

Subtracting proves (12)–(13). For every fixed $r > 3/2$, equation (11) and $q(r) > 1/3$ already imply eventual strict suppression under both rules. Thus $(1, 3/2)$ is exactly the asymptotic fixed-fitness amplification interval of $\{G_N\}$.

Remark 4 (Finite versus asymptotic claims). For every fixed $r \in (1, 3/2)$, the proof gives strict inequalities for every finite $N \geq N_0(r)$, but it does not provide one N_0 uniform over the open interval. At $r = 3/2$ and at each fixed $r > 3/2$, it gives eventual strict suppression for this family. It makes no assertion about all graphs at the endpoint.

11 Audited structural limits of related portal mechanisms

The center–triangle graph is a rare-coupling construction. Several broader portal–blade mechanisms were audited separately to test whether their establishment phase could cross $3/2$. The following class results are useful structural context, but none is a universal graph theorem and none is needed for Theorem 2.

1. **Fixed finite incidence rank without direct portal edges.** Take a fixed number Q of portals, a fixed number T of blade types with positive limiting proportions, unit blade edges, and portal–blade weights λ_{at}/s , with fixed positive incidence data. With no portal–portal edges, the exact stopped clean-blade trace has Bd and dB establishment bounds α_B, α_D satisfying

$$\alpha_B + \alpha_D < 2(1 - 1/r), \quad 3/2 \leq r \leq 2.$$

For $r \geq 2$, the dB entrance factor alone completes the exclusion. Thus this fixed-parameter class cannot simultaneously amplify at any fixed $r \geq 3/2$. Growing or singular incidence data and direct portal edges are outside the theorem.

2. **A direct two-portal scalar class.** For two portals, one blade type, arbitrary unequal positive portal loads, and an arbitrary positive direct portal edge, the exact labelled portal episode yields PGF margins T_B, T_D . An exact Bernstein certificate proves

$$T_B + \frac{81}{200}T_D < 0, \quad 3/2 \leq r \leq 2.$$

Hence both establishment bounds cannot beat the complete-graph limit in this class; $r \geq 2$ again follows from dB entrance. Three or more direct portals, higher-rank direct incidence, and singular parameter scales are not covered.

3. **Diffuse growing regular portal networks.** Let $Q_s \rightarrow \infty$, $Q_s = o(s)$, give each portal the same total portal-network load, assume the maximum individual portal edge tends to zero, and impose portal-exchangeable blade incidence. The exact labelled trace converges to a scalar birth–death immigration episode. Its exact fixed-point test proves a no-overlap theorem for every $r > 1$: if the total portal degree is below its Bd threshold, Bd establishment is too small; at or above that threshold, dB establishment is too small. This covers complete portal networks and degree-diverging regular expanders. Fixed-degree portal graphs, portal-dependent incidence, mesoscopic portal classes, and singular load scaling remain outside its scope.

Each statement concerns an establishment upper bound derived from a stopped trace; no inference that establishment automatically implies fixation is used. Collectively these results rule out specific natural extensions of the construction, but they do not prove $R_{\text{sim}} \leq 3/2$ or any other universal upper bound.

12 The normalized product and balanced mean are not separators

The endpoint obstruction, if true, cannot be obtained from either of two natural stronger inequalities. The following exact finite computation is included because it sharply separates those failed inequalities from the still-open disjunctive statement.

Proposition 5 (Exact endpoint counterexample to two stronger separators). *Let H be the unweighted graph obtained from K_{32} by distinguishing one clique vertex and adjoining four degree-one leaves to that vertex. At $r = 3/2$, put*

$$x = \frac{\rho_{\text{Bd}}(H, 3/2)}{\rho_{\text{Bd}}(K_{36}, 3/2)}, \quad y = \frac{\rho_{\text{dB}}(H, 3/2)}{\rho_{\text{dB}}(K_{36}, 3/2)}.$$

Then, exactly,

$$x > \frac{5609}{5000} > 1, \quad \frac{223}{250} < y < 1.$$

Consequently

$$xy > \frac{1250807}{1250000} > 1, \quad \frac{x+y}{2} > \frac{10069}{10000} > 1.$$

Thus both the universal normalized-product inequality and the balanced normalized-arithmetic inequality are false. The graph H is not a simultaneous amplifier at the endpoint because $y < 1$.

Proof. Write $c = 31$ for the ordinary clique vertices and $m = 4$ for the leaves. The automorphism group $S_c \times S_m$ preserves both update rules and acts transitively within each configuration fiber indexed by

$$(h, i, j) \in \{0, 1\} \times \{0, \dots, c\} \times \{0, \dots, m\},$$

where h is the type of the distinguished hub and i, j are the mutant counts in the other two classes. The total probability of every target fiber depends only on (h, i, j) , proving strong lumpability. Removing the two absorbing fibers leaves 318 transient states.

For each rule, insert the unweighted degrees $c + m, c, 1$ in (1)–(4). Let M_U be the resulting rational Dirichlet matrix after deleting self-loops, let b_U collect the rates into fixation, and solve

$$M_U H_U = b_U.$$

Uniform singleton initialization is exactly

$$\rho_U(H, 3/2) = \frac{H_U(1, 0, 0) + cH_U(0, 1, 0) + mH_U(0, 0, 1)}{36}.$$

The certificate solves both systems over $\mathbb{Q}$, substitutes the answer back into all 636 harmonic equations, and compares with the exact baselines (5)–(6). It obtains

$$x = 1.121822899272823\dots, \quad y = 0.892002982408856\dots,$$

where the displayed decimals are only annotations of exact rational numbers. Direct rational comparison gives the two small bounds in the statement. Their product is $1250807/1250000$, and their sum is $10069/5000$, proving the claimed strict signs without floating-point arithmetic. A second implementation rebuilds the transitions from labelled replacement events and independently solves the same two rational systems. $\square$

The exact affine crossing for this graph is

$$\lambda_0 = \frac{1-y}{x-y} = 0.469920183876266\dots$$

Hence $\lambda x + (1-\lambda)y > 1$ precisely when $\lambda > \lambda_0$. This calculation rules out the product and the balanced choice $\lambda = 1/2$; it neither rules out every affine separator nor proves the desired disjunction $x \leq 1$ or $y \leq 1$.

13 A growing product violation and the sharp affine coefficient

The finite violation above is not a conditioning artifact or a vanishing finite-size effect. Its growing version also identifies the largest Bd coefficient that a universal convex affine separator could have.

Theorem 6 (Growing endpoint product violation). *Let J_m be the unweighted graph consisting of K_{8m+1} , with one clique vertex distinguished as a hub, together with m degree-one leaves adjacent to that hub. At $r = 3/2$,*

$$\rho_{\text{Bd}}(J_m, 3/2) \longrightarrow \frac{32}{81}, \quad \rho_{\text{dB}}(J_m, 3/2) \longrightarrow \frac{8}{27}.$$

Consequently the normalized Bd and dB probabilities tend to $32/27$ and $8/9$, respectively, and their product tends to

$$\frac{256}{243} > 1.$$

The family is nevertheless asymptotically dB-suppressing.

Proof. More generally, let the number c_m of ordinary clique vertices and the number m of leaves tend to infinity with

$$\frac{m}{c_m + m} \longrightarrow \alpha \in (0, 1), \quad a := \frac{1 - \alpha}{\alpha}.$$

As in Proposition 5, the exact lumped state is (h, i, j) . We first record the post-establishment step that distinguishes the argument from a branching upper bound. If $k_m \rightarrow \infty$ and $k_m = o(m)$, then, under either update rule,

$$\inf_{h,j} \Pr_{(h,k_m,j)}(\text{fixation}) \longrightarrow 1. \quad (54)$$

Indeed, until i reaches $(1 - \delta)c_m$, the conditional up/down ratio of the ordinary-mutant count is $r - o(1) > 1$. In the upper strip, with $R = c_m - i$, the ratio of rates decreasing and increasing R is bounded below by

$$\frac{r(1 - 2\delta)R}{R + 1}.$$

Choosing δ with $r(1 - 2\delta) > 1$ gives exponentially long confinement near $R = 0$ by the standard exponential gambler-ruin supermartingale. More precisely, for every fixed A , the probability of escaping through $R = 2\delta c_m$ before time m^A is at most $C_A m^{A+1} e^{-\gamma_A m} = o(1)$, while R returns to a fixed strip $R \leq R_0$ in logarithmic mean time.

For Bd, observe the leaf process during these recurrent $R \leq R_0$ epochs. When $j = O(1)$, one hub excursion has probability $O(m^{-1})$ of one leaf change and $O(m^{-2})$ of two. On slow time t/m this converges to a linear leaf process with positive core immigration and per-particle birth/death ratio $r^2 + o(1) > 1$. Repeated immigrant families therefore hit $m^{2/3}$ in polynomial time with probability $1 - o(1)$. Between $m^{2/3}$ and εm , exact hub-excursion averaging gives leaf up/down ratio at least $r^2/(1 + 2\delta) > 1$, so return to zero before reaching εm has exponentially small probability. For leaf density $y = j/m$ bounded away from zero and one, the two-state hub occupation law gives on slow time $\tau = t/m$

$$\frac{dy}{d\tau} = \frac{y(1 - y)(r^2 - 1)}{(a + 1)\{1 + (r - 1)y\}} + o(1),$$

while the martingale quadratic variation is $O(m^{-1})$. For bounded leaf deficit $L = m - j$, the limiting rates are $rL/(a + 1)$ downward and $L/\{r(a + 1)\}$ upward. The same uniform embedded bias bridges $y = 1 - \varepsilon$ to $L = O(1)$ with exponentially small reversal probability. These block estimates carry

the leaf density to one and leave a subcritical deficit chain. During visits to $L = 0$, the hub is resident for only an $O(m^{-2})$ fraction of time when $R = O(1)$, while the R -rates converge to death rR and birth R . Polynomially many such visits therefore reach $L = R = 0$ and activate the hub before the exponentially unlikely core escape.

For dB, in the upper core strip the hub has a uniformly positive activation rate and a uniformly bounded deactivation rate. Choose C sufficiently large for both coupon collection and R -extinction. Each activation has probability at least m^{-C} of lasting $C \log m$, so a polynomial number of trials produces such an interval with probability $1 - o(1)$. During it, the leaf coupon process converts all leaves with probability $1 - o(1)$ and the $h = 1$ version of the R comparison is a subcritical linear birth–death chain which reaches zero. All stages take polynomial time, whereas core escape is exponentially unlikely. This proves (54), uniformly in the initial hub and leaf states.

From an ordinary singleton, stop when i first hits 0 or k_m . While the noncore mutant count is bounded, the early i -chain has rates $ri\{1 + o(1)\}$ and $i\{1 + o(1)\}$, and the accumulated probability of a hub or leaf event before this stop is $o(1)$. This follows by truncating first at a fixed ordinary-mutant level, taking $m \rightarrow \infty$, and then letting the fixed level grow. Equation (54) therefore turns the linear birth–death establishment probability into actual fixation:

$$u_C \rightarrow p := 1 - \frac{1}{r}. \quad (55)$$

For dB, an initial mutant leaf dies at rate one on the time scale obtained by multiplying transition probabilities by the population size, while its hub activation rate is $O(m^{-1})$. Hence $u_L^{\text{dB}} \rightarrow 0$.

For Bd, start from one mutant leaf and use the exact statewise time change which removes the total-fitness denominator. On slow time t/m , one leaf has limiting rates

$$\lambda = r^2\alpha, \quad \mu = \alpha, \quad \kappa = (1 - \alpha)r(r - 1) \quad (56)$$

for producing a second leaf, dying, and producing a core family that establishes. Stopping at a fixed leaf level or a mesoscopic core seed makes the convergence a finite-path calculation; failed core families have an exponentially decaying duration tail. If q is extinction before a successful core mark, branching independence gives

$$\lambda q^2 - (\lambda + \mu + \kappa)q + \mu = 0.$$

To remove the fixed leaf cutoff, let J be that cutoff in the limiting killed chain. Its probability of hitting J before extinction or a core mark tends to zero as $J \rightarrow \infty$: after adjoining the mark as killing, the no-mark process is a subcritical Doob transform. Taking first $m \rightarrow \infty$ and then $J \rightarrow \infty$ therefore identifies the full fixation limit with $1 - q$, rather than merely a stopped establishment probability. At $r = 3/2$, the root in $(0, 1)$ yields

$$\ell(\alpha) := 1 - q = \frac{8\alpha - 3 + \sqrt{9 + 60\alpha - 44\alpha^2}}{18\alpha}. \quad (57)$$

Thus $u_L^{\text{Bd}} \rightarrow \ell(\alpha)$. Uniform singleton initialization and the vanishing hub proportion give the normalized limits

$$x_\infty(\alpha) = 1 - \alpha + 3\alpha\ell(\alpha), \quad y_\infty(\alpha) = 1 - \alpha. \quad (58)$$

For $\alpha = 1/9$, equation (57) gives $\ell = 8/9$, hence $x_\infty = 32/27$ and $y_\infty = 8/9$. $\square$

Corollary 7 (Sharpness of the one-third coefficient). *Let $n = |V(G)|$. Suppose that for a fixed $\theta \in [0, 1]$ the convex affine inequality*

$$\theta \frac{\rho_{\text{Bd}}(G, 3/2)}{\rho_{\text{Bd}}(K_n, 3/2)} + (1 - \theta) \frac{\rho_{\text{dB}}(G, 3/2)}{\rho_{\text{dB}}(K_n, 3/2)} \leq 1 \quad (59)$$

holds for every finite connected undirected weighted graph. Then $\theta \leq 1/3$.

Proof. The affine crossing of (58) is

$$\theta_0(\alpha) = \frac{1 - y_\infty}{x_\infty - y_\infty} = \frac{1}{3\ell(\alpha)}.$$

Rationalizing (57) gives $\ell(\alpha) \rightarrow 1$ as $\alpha \downarrow 0$. Hence every $\theta > 1/3$ is violated by first fixing a sufficiently small rational α and then taking m large. At the sharp candidate itself the limiting slack is exactly

$$1 - \frac{x_\infty + 2y_\infty}{3} = \alpha\{1 - \ell(\alpha)\} > 0.$$

□

At the opposite end, an exact seven-vertex rational dB amplifier has normalized ratios $x_- < 1 < y_-$ and affine crossing

$$\theta_- := \frac{y_- - 1}{y_- - x_-} = 0.088542283991193063\dots > \frac{177}{2000}.$$

It follows by direct substitution that every universal convex affine separator must have

$$\theta_- \leq \theta \leq \frac{1}{3}.$$

Both endpoints of this necessary window are exact rational/algebraic certificates in the replay package; no coefficient in the window is currently proved universal.

The extremal affine candidate is therefore

$$\frac{1}{3} \frac{\rho_{\text{Bd}}(G, 3/2)}{\rho_{\text{Bd}}(K_n, 3/2)} + \frac{2}{3} \frac{\rho_{\text{dB}}(G, 3/2)}{\rho_{\text{dB}}(K_n, 3/2)} \leq 1. \quad (60)$$

It is universal if true, since two ratios exceeding one would violate it. Exact six-state elimination proves (60) for every positively weighted triangle: after clearing a denominator with 127 positive coefficients, the numerator is a 24-term positive sum of permutations of $a^i b^j c^k (a-b)^2$, with equality only at $a = b = c$. The full coefficient certificate is part of the replay package. For arbitrary order, (60) remains open.

With an unbounded number of leaves, the clique-pendant mechanism cannot be repaired by assigning arbitrary weights to them.

Proposition 8 (Uniform weighted clique-pendant obstruction). *Fix $r > 1$. Let $W(c; w_1, \dots, w_m)$ consist of the unit clique K_{c+1} with hub H , and attach leaf ℓ only to H with an arbitrary weight $w_\ell > 0$. There is a constant $A_r < \infty$, independent of c, m and all the weights, such that, for all sufficiently large c ,*

$$\frac{\rho_{\text{dB}}(W(c; w_1, \dots, w_m), r)}{\rho_{\text{dB}}(K_{c+m+1}, r)} \leq \frac{c + A_r}{c + m}. \quad (61)$$

Consequently every such graph sequence with $m \rightarrow \infty$ is eventually dB-suppressing, for every choice and scaling of the positive leaf weights.

Proof. Put $p = 1 - 1/r$, $W = \sum_\ell w_\ell$, and start with one mutant ordinary vertex. Until either the hub activates or the ordinary-mutant count i hits zero, the dB rates after a common time change are

$$b_i = \frac{(c-i)ri}{c + (r-1)i}, \quad d_i = \frac{i(c-i+1)}{c + (r-1)(i-1)}, \quad a_i = \frac{ri}{c + W + (r-1)i}.$$

Stop at 0 or $k = \lfloor c/2 \rfloor$. For fixed $1 < z < s < r$ and all sufficiently large c , the embedded up/down ratio lies in $[s, r]$ and $a_i/d_i \leq r(r+1)/c$. The potential

$$V(i) = \frac{z^{-i}}{1 - (s/z + z)/(s+1)}$$

bounds the expected sum of the future-extinction probabilities s^{-i} along this embedded path. Granting fixation after a hub activation or after hitting k therefore gives, uniformly in all leaf weights,

$$u_C \leq \frac{p}{1 - r^{-k}} + \frac{C_r}{c}$$

for a constant C_r .

From a mutant leaf ℓ , the only type-changing events before extinction or hub activation give exact activation-first probability

$$\frac{rw_\ell}{c + W + (2r - 1)w_\ell} \leq \frac{rw_\ell}{c + W}.$$

Even granting fixation after activation, summing over all leaves gives $\sum_\ell u_{L_\ell} \leq r$. Grant fixation from the initial hub, average the $c + m + 1$ singleton starts, and divide by (6). The exponentially small factor $cr^{-\lfloor c/2 \rfloor} / (1 - r^{-\lfloor c/2 \rfloor})$ is bounded and can be absorbed into A_r , which proves (61). For an arbitrary sequence with $m \rightarrow \infty$, split its indices into $c \geq c_0(r)$ and $c < c_0(r)$. On the first subsequence, (61) is below one once $m > A_r$. On the second, the total contribution of the hub and ordinary singleton starts is at most $c_0(r)$, while the total leaf contribution is at most r ; division by the complete baseline therefore tends uniformly to zero. This proves the eventual conclusion even when the two cases alternate. $\square$

14 Exact and independent verification

The proof package contains definition-level transition builders and symbolic certificates. The discovery formulas are not imported as assumptions: the triangle programs rebuild the subset chains from (1)–(4), and the quotient audit compares a full labelled chain against the symmetry-reduced transition map state by state. All asserted identities use exact rational or symbolic arithmetic.

Verifier	Independent task	Arithmetic / expected status
Triangle module	Builds both six-state triangle chains; checks (16)–(23), scale-window factors, center reverse ratios, and endpoint coefficients.	Symbolic rational functions; PASS.
Independent triangle	Independently rebuilds the triangle subset systems and rare-migration rate identities from the atomic rules.	Symbolic rational functions; PASS.
Full-state quotient	Enumerates every labelled transient state for two small rational instances and compares its aggregate transitions with the quotient chain under both rules.	Exact fractions; PASS.
Fixed-rank portal	Regenerates the fixed-rank no-direct-edge separator and its Bernstein cover.	Exact integer/rational certificate; PASS.
Labelled direct trace	Rebuilds every nonempty labelled portal subset for a rational $Q = 3, T = 2$ instance and checks first-step row balance and the reduced maps.	Exact fractions and rational functions; PASS.
Direct two-portal	Regenerates the direct two-portal separator and audits strictness on all physical boundary faces.	Exact Bernstein certificate; PASS.
Diffuse portal limit	Checks the growing diffuse portal episode and the scalar no-overlap factorizations.	Symbolic exact arithmetic; PASS.
Endpoint product witness	Solves both 318-state quotient chains over $\mathbb{Q}$, checks every harmonic equation, and certifies the strict bounds in Proposition 5.	Exact rational arithmetic; PASS.
Independent endpoint witness	Rebuilds the quotient rows from labelled replacement events and reproduces both fixation fractions with a separate exact solver.	Exact rational arithmetic; PASS.
Growing clique-pendant	Checks the general core-to-leaf averaged rates, the killed branching polynomial, the limits in Theorem 6, and labelled lumping independently.	Exact fractions and symbolic identities; PASS.
One-third Green/triangle audit	Rebuilds the exact Green–Poisson identity on ten hostile graphs and certifies the positive 24-term weighted-triangle factorization.	Exact rational and polynomial arithmetic; PASS.
Arbitrary pendant weights	Checks common-weight labelled lumping, the symbolic embedded core ratios, and the arbitrary-weight leaf-activation bound in Proposition 8.	Exact fractions and symbolic identities; PASS.

Table 1: Independent verification paths included in the replay package. The recorded replay log reports the actual result of each invocation.

The one-command replay script also records interpreter and dependency versions. The build script compiles this source deterministically into the paper folder. Computational checks validate the finite algebra and the implementation; the universal-in- N conclusion is supplied by the analytic error bounds above, not by finite testing.

15 Conclusion and open problem

The explicit rational family (8) proves $R_{\text{sim}} \geq 3/2$. Its mechanism separates the local fixation probability $1/3$ from a nearly deterministic two-stage macro takeover. The rare-edge lemma makes this separation quantitative through all excursions, and the endpoint expansion determines the family’s exact interval.

Proposition 5 also settles two tempting stronger endpoint conjectures in the negative. In particular, a proof of endpoint impossibility cannot proceed merely by bounding the normalized product or its balanced arithmetic relaxation. The counterexample does not amplify dB, so it leaves the actual disjunctive obstruction untouched. Theorem 6 shows that the product failure persists by a fixed margin in growing populations. Corollary 7 then identifies the maximal possible Bd coefficient in any universal convex affine separator: Bd can receive weight at most $1/3$. The resulting one-third inequality is proved for all weighted triangles but remains open in general.

The central unresolved question is whether another fitness-independent family can amplify both update rules at or beyond $r = 3/2$. Equivalently, the universal upper bound and the exact value of R_{sim} remain open. The family-specific scale window and the portal class obstructions in Section 11 identify mechanisms that do not cross the endpoint. Proposition 8 additionally closes every growing clique-pendant construction with an unbounded number of arbitrarily weighted leaves. None of these class theorems is a universal separator.

Scope. This paper proves the lower bound $R_{\text{sim}} \geq 3/2$ and the exact interval of one explicit family. A universal upper bound and the exact value of R_{sim} remain **open**.

Reproducibility and disclosure

All graph weights, transition equations, proof constants, and exact certificates used here are included in the accompanying local source package. No numerical fixation plot is used as proof. The manuscript and verifier package were prepared with AI assistance under human-directed independent research; all mathematical claims stated as proved were subjected to exact replay and a separate hostile audit. Any public distribution of this manuscript is a preprint, not a claim of peer-reviewed publication or journal submission. No broad priority claim is made.

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